DPP

Analysis of the algorithm:
“Class-based graph anonymization for social network data”

DATASET

The dataset that I have used for this analysis is from a reviews community of films, starting from it I crafted the interaction graph: a bipartite graph over the sets V and I, where:

- V are users
- I are films

If there is an edge between a user and a film, it means that the user reviewed the film.

Each user has as attributes id, number of ratings and age.

Privacy-utility level

Experimental setup

For this part of the analysis I decided to test the anonymized graphs with different queries, comparing the result with the ones obtained from the original graph. These are the queries:

- How many users with num_ratings > min_ratings (= 40) rated a movie from target_genre (= Documentary)
- Counts users with exactly N (= 25) ratings
- Counts user pairs who share at least N (= 5) movies.

Experimental setup

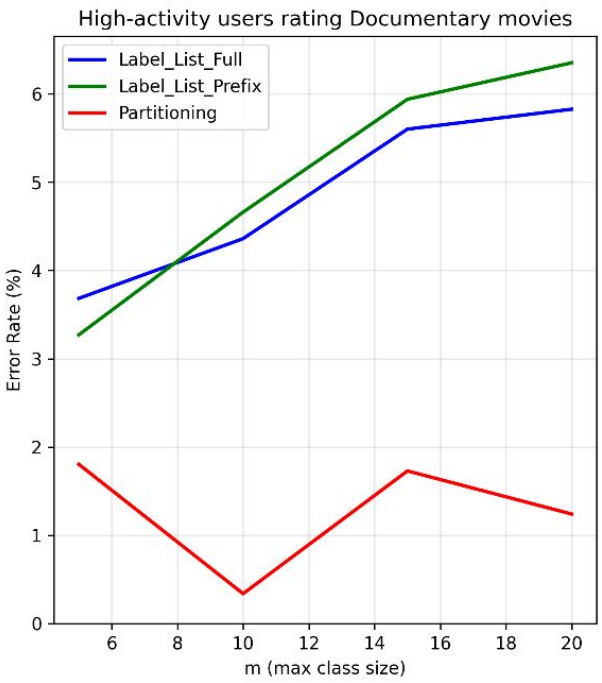
For each anonymization algorithm (Label List Full pattern, Label List Prefix pattern and Partitioning), I have tested the query with different parameters:

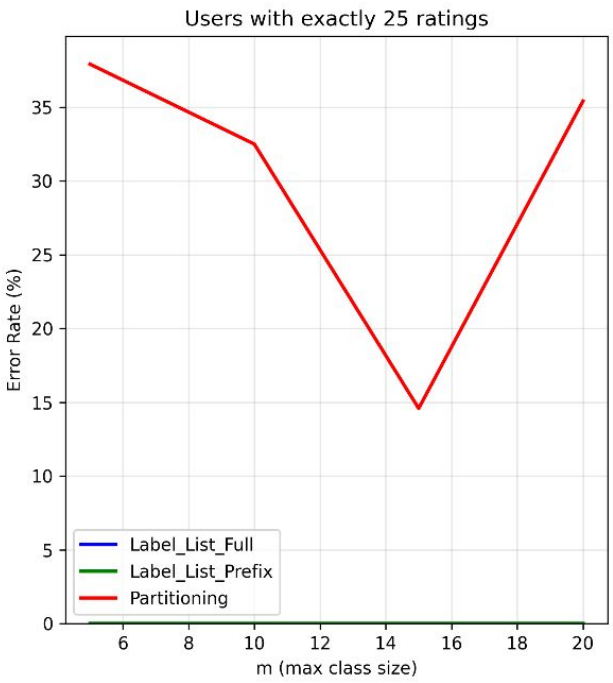
- **m** = [5-10-15-20]
- with **k** as the 70%-90% of m values (for prefix pattern)
- for two distinct **sort_orders** attributes used for creating classes of nodes: ‘num_ratings’, ‘age’

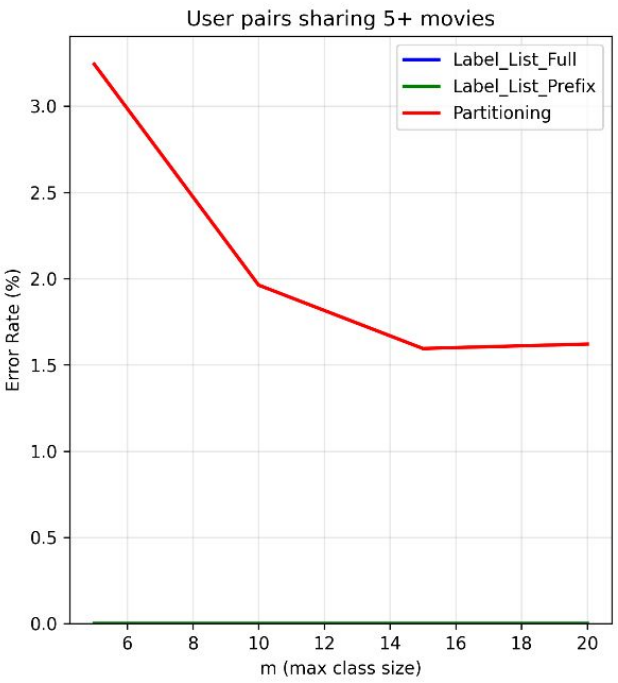
Experimental setup

- For reconstructing an interaction graph after the anonymization process I randomly **sample** a **graph** that is **consistent with the anonymized data**, and perform the analysis on this graph.
- Each query can be evaluated over the anonymized graph repeating this several times (10 in my experiment), computing the average of the different samples and obtaining an **“expected” answer** to the query
- **Utility** refers to **how much the “expected” answers** given by queries differ from the results on the original graph

Obtained Results







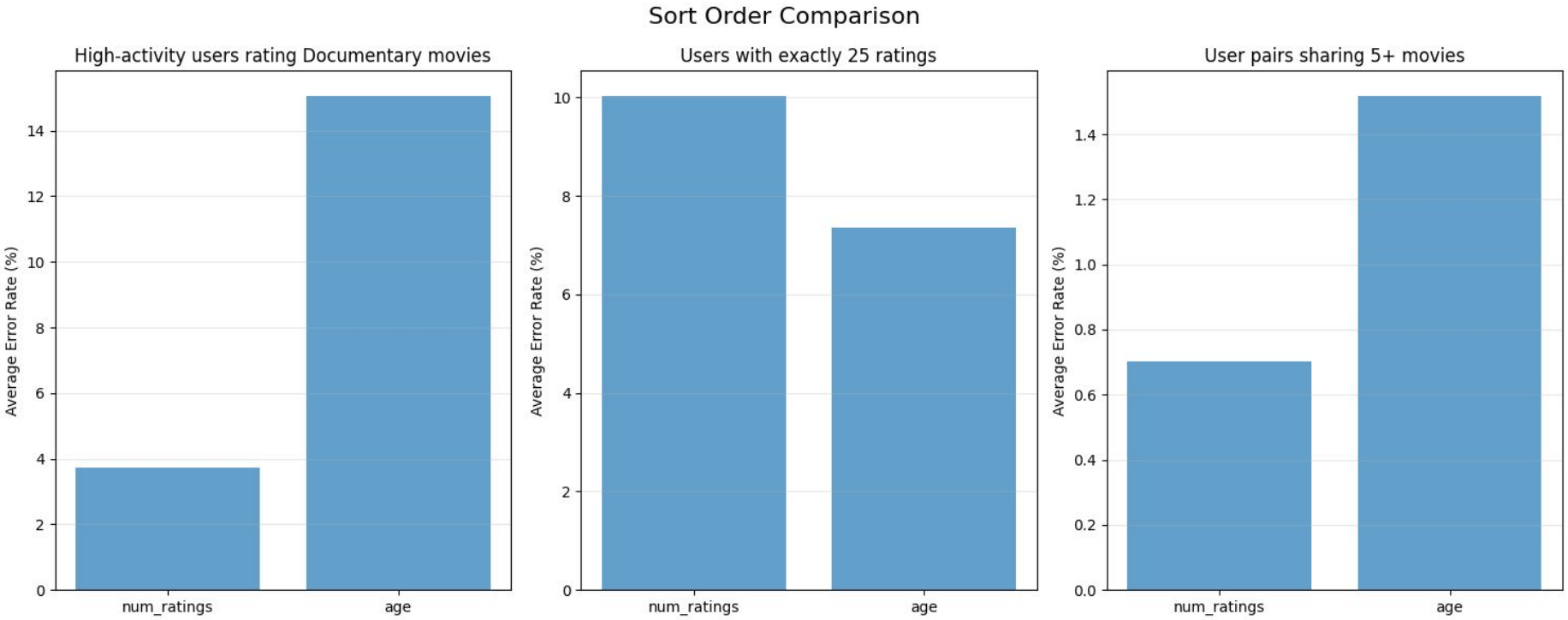
Obtained Results

- The **first query** is mainly **focused on attributes** associated to users (num_ratings), this is the reason why **partitioning algorithm perform better** than the label list algorithms because the **partitioning** method **alters the structure of the graph** but **not the entity associate to a vertex**
- On the other hand **this feature is a problem** for partitioning method in the **second and third queries**, because this two are queries that are mainly focused on the interactions set of each node, the label list methods don't change the interaction set of the users, for this reason the error is null

Obtained Results

- In the first query, we can observe that **the error increase with the increasing of m**, this because with **m** we can **regulate** the **tradeoff between privacy and utility**: **low m** means small label list for each vertex, hence **small privacy** but **results more similar to the original** ones (**utility**); it is the **opposite with the increasing of m**

Obtained Results



Obtained Results

- A last point is the **importance of the sorting attribute** in **queries** that mainly **focus on a specific attribute** of a user rather than on his interactions (first query): as we can observe if we choose to **sort by that specific attribute** we will **improve the utility**, minimizing the error

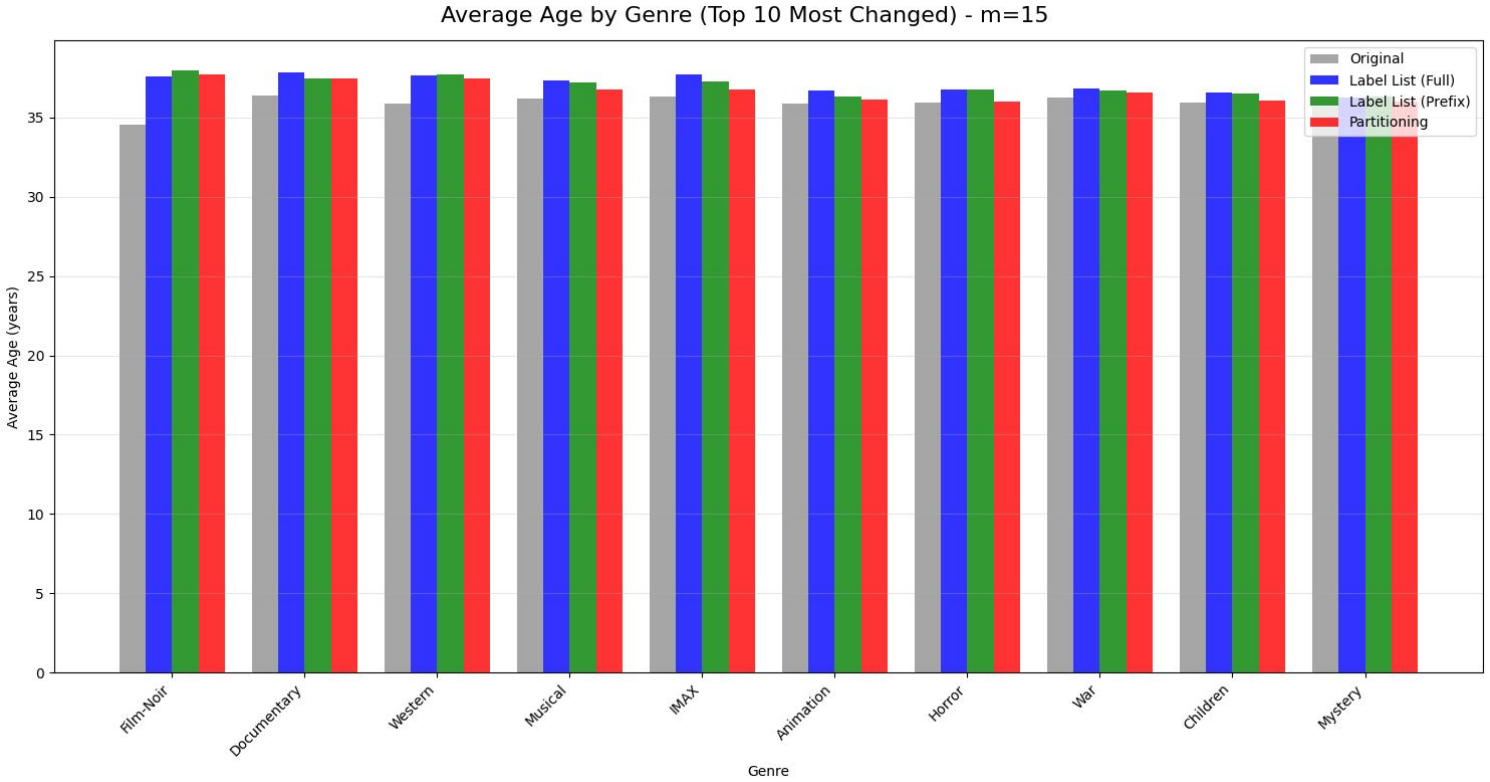
Statistical information before/after

Experimental setup

To understand how much the different anonymization techniques have changed the statistical distribution, I chose to analyze **for each movie genre** the **average age of the people who reviewed** it, these are the parameters used:

- **m =** [5, 10, 15, 20, 25, 30]
- **k =** 70% of m (for prefix pattern)
- sorting by ‘**age**’

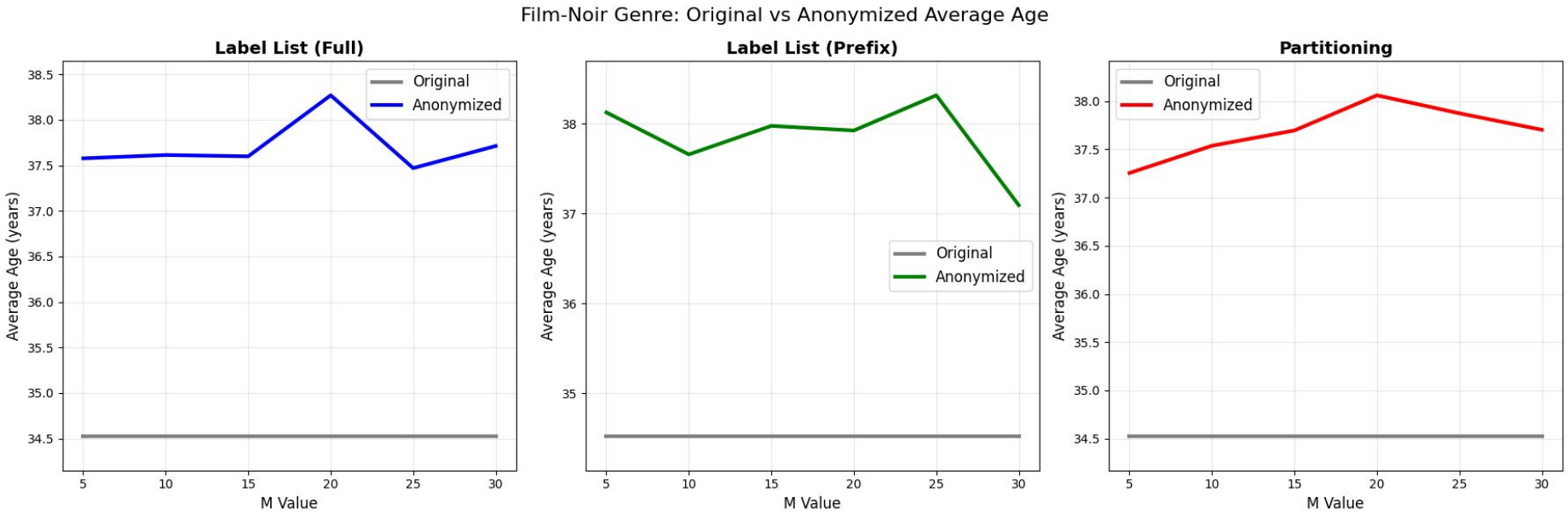
Obtained Results



Obtained Results

- Looking at the following plot, we can see that the **statistical distribution has not changed significantly** from any of the different anonymization techniques.
- In fact, the **top 10 genres** that have received the greatest change are shown, and in the worst case (Film Noir), we can consider the maximum difference from the original genre mean (approximately 4 years) to be relatively small.
- And we can see how in the **remaining genres the differences are increasingly smaller**, becoming almost insignificant.

Obtained Results



Obtained Results

- Even if we increase the m value, increasing the probability to deviate more from the original distribution, we can observe that this doesn’t happen so we can conclude that all the anonymization techniques guarantee a good level of utility

Efficiency Analysis

Experimental setup

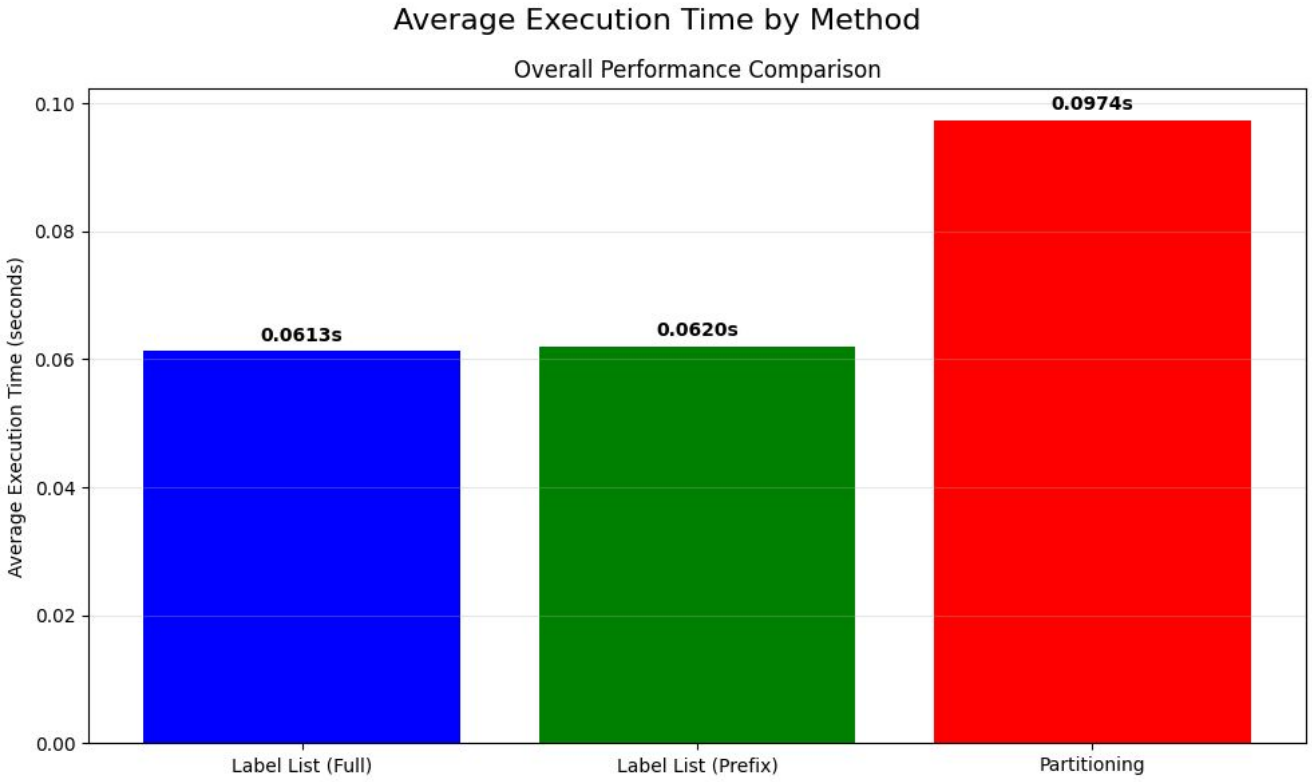
I chose to measure the **runtime** of the three different **anonymization techniques** on the original movies graph that has:

- 610 users (V)
- 100836 interactions (I)

Using different parameters:

- m = [5, 10, 15, 20, 25, 30]
- k = 70% of m (for prefix pattern)
- sorting by ‘age’ (for ‘num_ratings’ the results are the same)

Obtained Results



Obtained Results

- As we can see, all anonymization techniques have small runtimes: for each method, we have runtimes < 0.1 seconds
- We can also see that partitioning is less efficient than the other two because it has to build a more complex data structure that represents the partitioned anonymized graph